SEONGLAE CHO

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Senior AI Engineer at Holistic AI and mechanistic-interpretability researcher. First-author ICML 2026 main publication; agent runtime safety through steering, probe-as-tool, and realtime agent monitoring, taken from research to production.

SELECTED PUBLICATIONS

- Cho, S., Wu, Z., & Koshiyama, A. (2025). CorrSteer: Generation-Time LLM Steering via Correlated Sparse Autoencoder Features. **ICML 2026 main.** +27.2% HarmBench (108 samples), +3.3% MMLU. demo
- Cho, S., Wu, Z., & Koshiyama, A. (2026). Control Reinforcement Learning: Interpretable Token-Level Steering of LLMs via Sparse Autoencoder Features. ICLR 2026 Workshop on Principled Design for Trustworthy AI.
- Cho, S. et al. (2026). PaaT (Probe as a Tool): self-regulating agents via a callable activation probe. ICML 2026 FAGen Workshop. Probe AUROC 0.93–1.00 across 11 risk heads.
- Cho, S. et al. (2026). AgentRoom: Concurrent Multi-Agent Coding in a CRDT-Backed Shared Workspace. ICML 2026 FAGen Workshop. 71% of multi-agent gain attributed to the coordination layer.
- Cho, S., Wu, Z. et al. (2026). Automata from Agent Traces: Failure and Next-Step Prediction. ICML 2026 AIWILD Workshop. 0.975 AUROC failure detection.

PUBLICATIONS

- Cho, S., Wu, Z. et al. (2026). The Confidence Manifold: Geometric Structure of Correctness Representations in Language Models. ICLR 2026 Workshop on Agentic AI in the Wild. Internal probes beat output-based detection by 23–33pp AUROC across 9 LLMs.
- Cho, S., Wu, Z. et al. (2026). Are Single-Token Sparse Autoencoder Features Irreplaceable? The Impact of Training Methodology. EMNLP 2026 under review.
- Cho, S. et al. (2025). FaithfulSAE: Towards Capturing Faithful Features with Sparse Autoencoders without External Datasets Dependency. ACL 2025 SRW. Outperformed external-data SAEs in 5/7 models.
- Wu, Z., Cho, S. et al. (2025). AgentGraph: Trace-to-Graph Platform for Interactive Analysis and Robustness Testing in Agentic AI Systems. AAAI 2026 Demo.
- Wu, Z., Cho, S.* et al. (2025). LibVulnWatch: A Deep Assessment Agent System and Leaderboard for Uncovering Hidden Vulnerabilities in Open-Source AI Libraries. ICML 2025 TAIG Workshop. demo
- Cho, S. et al. (2024). RTSUM: Relation Triple-based Interpretable Summarization with Multi-level Salience Visualization. NAACL 2024 System Demonstrations.

EXPERIENCE

HOLISTIC AI, London, UK · *Senior AI Engineer* — May 2025 – Present

- Built a trace-based AI agent evaluation framework with structured traces, quality criteria, and graph-based monitoring; a 300MB toxic-chat classifier beat gpt-oss-safeguard-20B on F1 (0.836 vs 0.799).
- Designed a trace-to-FSM reconstruction method reaching 0.904 runtime-monitoring F1 at 68% lower compute.
- Integrated ReBAC access control into the AI Governance platform with database-level policy enforcement.
- Promoted to Senior within one year of joining.

YONSEI UNIVERSITY, Data & Language Intelligence Lab, Seoul, South Korea · *Research Intern, Prof. Dongha Lee* — Mar 2023 – Jan 2024

- Interpretable-summarization research on LLMs; first-authored RTSUM (NAACL 2024 Demo).

KAKAO MOBILITY, Seoul, South Korea · *Software Engineer, Digital Twin Team* — Dec 2021 – Sep 2022

- Built Kubernetes infrastructure with Grafana alerting and isolated namespaces.
- Led a 3-engineer team delivering a 3D point-cloud processing service handling 10M+ point cloud frames in the autonomous-driving pipeline.
- Ported a C++ 3D-projection algorithm to Rust with Node.js N-API bindings for cross-platform deployment.

STRYX, Seoul, South Korea · *Software Engineer, Geospatial AI Startup* — Nov 2019 – Dec 2021

- Cut build time 70% by consolidating multiple repositories into a monorepo.
- Reduced the Docker image from 2GB to 180MB via multi-stage CI builds.

EDUCATION

University College London · *MSc in AI for Sustainable Development, Distinction* — Sep 2024 – Sep 2025

- Dissertation: Representation-Level Steering Improves Task Performance and Safety in LLMs. Supervisor: Prof. Philip Treleaven. Mentor: Zekun Wu.

Yonsei University · *BE in Computer Science* — Mar 2017 – Aug 2024

AWARDS

- **HERMES, 1st Place (£3,000), Holistic AI Hackathon** (Nov 2024) · cut biased, stereotypical text generation from 90% to 70% with an SAE-derived steering vector.
- **MBTIGPT, 1st Place (₩3,000,000), Yonsei GenAI Competition** (2023) · AI service with 1,000+ users, RAG over chat history with Redis and Faiss.

PROJECTS

- **OptimismBench** · multilingual benchmark for directional bias in LLM probability judgment across 6 languages and 16+ models. EMNLP 2026 under review. — Feb 2026 – Present
- **Stochastic Grokking** · grokking hinges on a few attention heads binding to variable slots; transplanting these heads rescues 6/6 failed seeds. NeurIPS 2026 under review. — Jan 2026 – Present
- **RouterRL** · trained only MoE router parameters with dense layer-wise probe rewards, matching SFT where sparse binary RL fails to learn. — Feb 2026 – Mar 2026
- **ReSRer** · retrieval-augmented generation pipeline indexing 21M documents into a Milvus vector database. — Sep 2023 – Jan 2024
- **Texonom** · Zettelkasten knowledge system; ANN retrieval over 30,000 pages embedded into Postgres pgvector. — Nov 2021 – Jun 2023

SKILLS

- **AI**: Python, PyTorch, Hugging Face, vLLM, ONNX, DDP/FSDP, PydanticAI, RAG, LangGraph, Agent SDK, Langfuse, LangSmith.
- **Service**: TypeScript, Node.js, Rust, React, Next.js, PostgreSQL, Redis, Faiss, Milvus, Vite.
- **Infra**: Kubernetes, Linux, Git, CI/CD, Docker, Apache Spark, Ansible, ETL, AWS, GCP, Hadoop, Terraform, SST.
- **Languages**: Python, TypeScript, Rust, C++.

SERVICE

- Reviewer, ICML 2026 (Silver Reviewer Award) · Reviewer, ACL Rolling Review (March 2026).
- Mandatory military service completed as an Industrial Technical Personnel.